



IX IMW
IBERIAN MODELLING WEEK

Route Optimization for a set of
Vendors

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1 Introdução

Brief description of the problem “challenge”:

A company whose vision is to be one of the world leaders in transport solutions, construction equipment and agricultural equipment has, over the last few years, invested in industrial mathematics with the aim of improving its efficiency in several dimensions.

The challenge proposed for this Iberian Modelling Week is based on determining the best partition of customers among the company’s sales staff and building the respective visit route. Each customer has a member of staff (sales staff) associated with it, who must carry out a visit with a certain frequency (depending on the type of customers). At this moment, there is already a partition of the customers with respect to the salesmen.

This challenge has two main goals:

1) Given the existing partition of customers, how to find an optimal route to visit customers considering the last time they were visited, the periodicity of the visits and the distances between customers.

2) Given the matrix of distances between customers, is there a more favourable partition of customers, taking into account the number of available salesmen?

This challenge was proposed by Professor Manuel Cruz, Departamento de Matemática, Instituto Superior de Engenharia do Porto (LEMA-ISEP-IPP), Porto, Portugal.

2 Territory Decision

Dividing the total customers area into territories is a strategy to deal with the complexity of the Vehicle Routing Problem (VRP). This approach involves solving smaller subproblems that are easier to solve and allows for resource reuse by serving territories on different days. Furthermore, this division into territories brings benefits for improving customers service. Salespeople who regularly visit the same set of customers are familiar with the region, resulting in shorter service and travel times (Schneider et al., 2014)[4]. Operating with territories is also simpler than managing the entire system at once, and in case of changes such as adding or removing a customer, only a portion of the global planning needs to be adjusted.

The challenge that has been posed is to improve and find an optimal solution for the problem described in the previous section, where we have 9 salespeople to serve or visit at least 500 customers in a city. In this context, we define the work hours per day for each salesperson as 5 hours, working 5 days per week, which allows for some flexibility for the salespeople and facilitates clustering. Considering a monthly timeframe, it is evident that each salesperson has 100 hours of work per month. With the objective of generating solutions for $k = 1, \dots, 9$, we aim to generate solutions for $k = 8, 9, 6, 5$, and so on, and answer the question “what is the best way to improve the distribution of salespeople to customers in a timely manner?”

In the first part of the Figure 1 illustrates the existing dataset or the information we

have in the customers database. In this regard, the second part of the same figure illustrates depicts each of the clusters per salesmen with the purpose of selecting one for modeling in the next section.

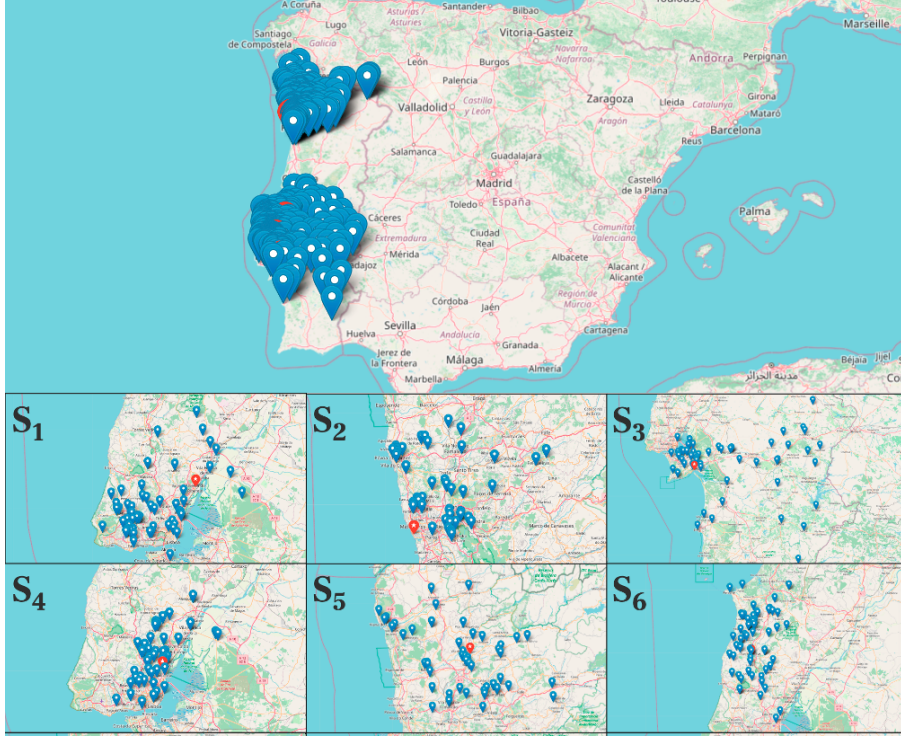


Figure 1: Information of the customers database and the clusters (S_n)

However, it is important to note that the quality of clustering influences the routing solution. An improper clustering assignment can result in longer travel times, leading to higher operational costs and poor customers service (Hadjiconstantinou and Giosa [1]).

3 Vehicle Routes Problem With Time Windows

The Vehicle Routing Problem with Time Windows (VRPTW) is a significant variant of the classic Vehicle Routing Problem (VRP) that arises in various real-world logistics and transportation scenarios. In the VRPTW, a fleet of vehicles is tasked with serving a set of customers, each with a specific demand, within a given time window. The objective is to find optimal routes for the vehicles, minimizing total travel distance or time, while adhering to the time constraints imposed by the customers' availability [Toth, P., & Vigo, D. (2002)][5].

The VRPTW presents several practical challenges due to the time window restrictions, which require careful scheduling and coordination to ensure timely deliveries and avoid customer dissatisfaction. The problem becomes more complex when considering multiple depots, heterogeneous vehicles, and various constraints such as capacity limitations and delivery priorities.

Over the years, the VRPTW has received significant attention from researchers and practitioners, leading to the development of various algorithms and optimization techniques to tackle the problem. These approaches include heuristic methods, metaheuristics, exact algorithms, and hybrid approaches that leverage advances in operations research, computer science, and transportation engineering.

This research work aims to explore and propose solutions for the VRPTW, investigating the heuristic approach and evaluating its performance of the model in which we found it. By addressing this issue, we seek to contribute to the improvement of transport and logistics operations, leading to more efficient and cost-effective delivery systems.

And in our specific case, we have p customers that are visited in a period of 30 days according to which we have to visit them with a frequency of f_i where $i = 1, \dots, p$. We have to consider the time of the working day, which amounts to 6 hours a day during trips between customers and their appointments. We are going to model our problem using linear programming with constraints. First of all, we will define our parameter that we will use in our problem.

Set	Description
C_i	customers
V	Vertices
$\#V$	$\sum f_i$
k	set of working-day [0...20]
Parameters	Description
f_i	Frequency of visit
d'_{ij}	Distance matrix times per customers
d_{ij}	Distance matrix vector
d_i	Last visited day
a_i	Appointment time

The data that we have is,

customers	Frequency	Last visit day
C_1	f_1	d_1
C_2	f_2	d_2
$\vdots$	$\vdots$	$\vdots$
C_p	f_p	d_p

Vertices, dimensions of our problem is,

$$V = C_1^1, C_1^2, \dots, C_1^{f_1}, C_2^1, \dots, C_p^{f_p} \implies V = \sum f_i \quad (1)$$

3.1 Little change to customers and distance matrix

$$\begin{matrix} & C_1 & C_2 & \dots & C_p \\ C_1 & \begin{pmatrix} 0 & d_{12} & \dots & d_{1p} \end{pmatrix} \\ C_2 & \begin{pmatrix} d_{21} & \ddots & & \vdots \end{pmatrix} \\ \vdots & \begin{pmatrix} \vdots & \vdots & \ddots & \vdots \end{pmatrix} \\ C_p & \begin{pmatrix} d_{p1} & \dots & \dots & 0 \end{pmatrix} \end{matrix} \rightarrow \begin{matrix} C_1^1 & \dots & C_1^{f_2} & \dots & C_2^1 & \dots & C_2^{f_2} & \dots & C_p & \dots & C_p^{f_p} \\ \begin{pmatrix} C_1^1 \\ \vdots \\ C_1^{f_2} \\ \vdots \\ C_2^1 \\ \vdots \\ C_2^{f_2} \\ \vdots \\ C_p \\ \vdots \\ C_p^{f_p} \end{pmatrix} & \begin{pmatrix} 0 & & & & & & & & & & \\ & 0 & & & & & & & & & \\ & & 0 & & & & & & & & \\ & & & 0 & & & & & & & \\ & & & & 0 & & & & & & \\ & & & & & 0 & & & & & \\ & & & & & & 0 & & & & \\ & & & & & & & 0 & & & \\ & & & & & & & & 0 & & \\ & & & & & & & & & 0 & \\ & & & & & & & & & & 0 \end{pmatrix} \end{matrix} \quad (2)$$

And, each block is:

$$\begin{matrix} & C_1 & \dots & C_p^{f_1} \\ C_1 & \begin{pmatrix} 0 & max & max \end{pmatrix} \\ \vdots & \begin{pmatrix} max & \ddots & max \end{pmatrix} \\ C_p^{f_1} & \begin{pmatrix} max & \dots & 0 \end{pmatrix} \end{matrix}$$

So we solve this problem by separating the customer i so much time like its frequency, so we change our problem of visiting f_i times in visiting each vertex only one time.

$$d_{i^*j} = max|D| + 1 \quad \text{and} \quad d_{ij^*} = max|D| + 1$$

$$\begin{aligned} i^* &= 1, \dots, p \\ j &= i^*, \dots, f_{i^*} \\ j^* &= 1, \dots, p \\ i &= j^*, \dots, f_{j^*} \end{aligned}$$

The others d_{ij} is the same that the customer it represents, this is because the new vertices are a copy of the first one.

3.2 Formulation

Our variables is:

$$\phi_{ijk}, \quad \text{with } i, j = 1, \dots, V(end), \quad k \text{ is days of the months} \quad (3)$$

$$\phi_{ijk} = \begin{cases} 0, & \text{cc} \\ 1, & \text{if day } k \text{ our works will go from } v_i \text{ to } v_j \end{cases}$$

The objective function of our problem is,

$$F.O. : \min L, \quad \text{where}$$

$$L : \sum \phi_{ijk} t_{ijk}$$

and t_{ijk} is a distance matrix based on time, and $\forall k \in T$.

1) Begin from D-pot, office departament, and we have to put another line and row in distance-matrix, with the distance to D-pot. We define like vector 0.

$$\sum_{j \in V \cup \{0\}} \phi_{0jk} = 1, \quad \text{fixed } k \forall k.$$

2) Contunity:

$$\sum_j \phi_{ilk} - \sum_j \phi_{ljk} = 0, \forall l \in V \quad \text{and} \quad \forall k \in T.$$

3) Dont go to himself:

$$\phi_{iik} = 0, \quad \forall i, \forall k.$$

4) Visit all the vertices:

$$\sum_{ijk} \phi_{ilk} = 1, \quad \forall i, j, k.$$

5) Time of work:

$$\sum \phi_{ijk} t_{ijk} + \phi_{ijk} d_j \leq 6 \text{hours/week}$$

6) Time Windows:

We have to avoid go from one copy of the client i to another. We also put in distance-matrix the maximum cost to avoid but now we have to do in constraints.

$$C_i^l, \quad \text{with } l = 1, \dots, f_i$$

$$\text{Fixed customer } i : \begin{cases} \sum \phi_{ijk} & = 1, \quad k \in [0, \frac{20}{f_i}] \\ \vdots & \vdots \\ \sum \phi_{(i+f_i-1)jk} & = k \in [\frac{(f_i-1) \times 20}{f_i}, 20] \end{cases} \quad j \notin [\sum_{p=1}^{i-1}; \sum_{p=1}^i]$$

7) Avoid Subtour:

With fixed k and S , $\forall S \in V, 1 \leq |S| \leq \sum(f_i - 1)$ A subtour is a travel without sense, because it has not contunuity between vectors.

For example $(0,1,0)(2,3,4,2)$

It pass all the constraints but it is not the path that we want. So we have to delete all the possibilities which make this situation.

If there is at least one conection in S ,

$$\sum_{i \in S} \sum_{j^* \in V \cup \{0\}} \phi_{j^*ik} \geq 1$$

Than, must be have one conection between S and its complementary. Clarify, if you go in S you have to go out, so you can't do loops.

$$\sum_{j \notin S} \sum_{i \in S} \phi_{ijk} \geq 1$$

So this is the constraint, if there is no conection in S , do nothing else it has to be one conection to complementary of S , at least.

$$\sum_{i \in S} \sum_{j^* \in V \cup \{0\}} \phi_{j^*ik} \leq |S| \sum_{j \notin S} \sum_{i \notin S} \phi_{ijk}$$

3.3 Heuristic

We have already do mathematic formulation but we know that the problems NP (no computable), because the order increase by exponential in the number of customers and their frequency. So we can apply some heuristic methods to solve the problem. This time we use OR-TOOLS and its program to solve (VRPTW)

3.4 Pratical implementation

For the implementation, we started making cluster of the customers and clasify by worker, so the problem we have to face 9 times is making a monthly route to one worker which start and end at D-pot each day and go through all customers, with all the constraints that we have explained previously.

We will solve with OR-TOOLS and we use matlab to make the graphic, each colour represent one routine.

So, this is an example with the information in ['Visitass-alunos.xlsx'](#)

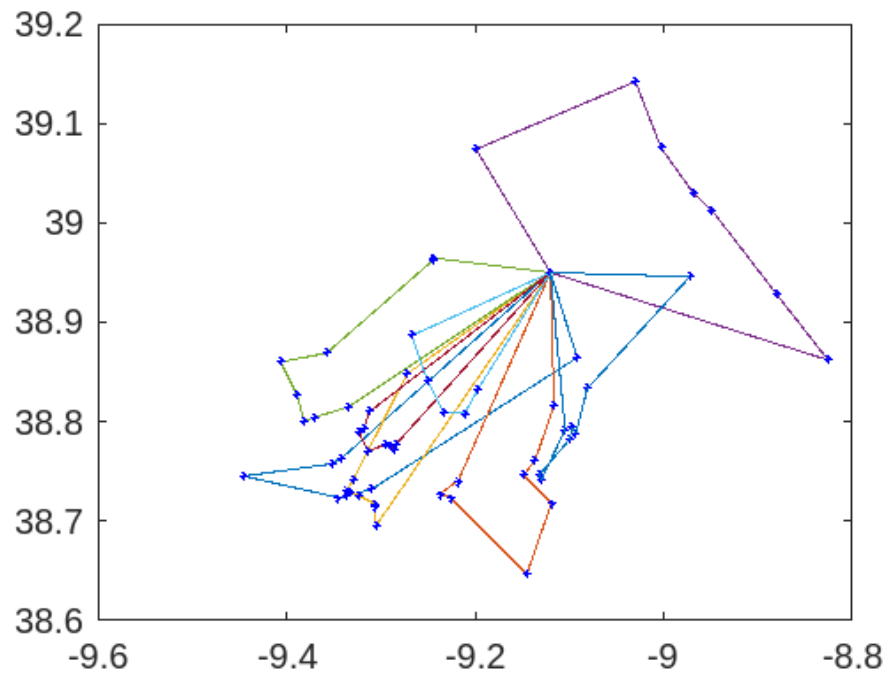


Figure 2: Worker route for one month

We began with geographic distance of all the customers, we do cluster we focus in only one worker each time and his customers, and we got the distance matrix for this worker. [haversine-distance.m](#).

We only consider frequency one this time, but we made a program to duplicate customers like we already say. [MatrizVector2.m](#).

3.4.1 Parameters

OR-TOOLS only require Matrix-distance, Number of vehicles, Time-Windows for each customer and Max time for each vehicle. We have to modify our problem using these parameters.

For example how do we add the time of meetings? It's simple we add to the Matrix-distance, we plus in each row (i) of the matrix the time meeting of the customer(i).

The Number of vehicles represent the numbers of days in the month available to work, so without weekends, is 20 days usually.

Time-Windows rely on frequency-visit in this case you have to visit one time each, so the time windows are [0,20] for all the customers. The Max timer for each vehicle represent hours of work-day we can use 6 hours. The problem that we found was you can not use decimals in matrix distance, so we use all the parameters measure in minutes so we don't lose any information. So 20 days to 120 hours to 7200 minutes and a day-work is 360 minutes.

This is the solution when you run the code for worker 1 on day 15:

Route for vehicle 15:

0 *Time*(0,0) → 14 *Time*(56,56) → 57 *Time*(88,88) → 51 *Time*(119,119)

→ 16 *Time*(151,151) → 48 *Time*(183,183) → 31 *Time*(214,214)

→ 50 *Time*(245,245) → 2 *Time*(276,276) → 0 *Time*(332,332)

Timeoftheroute : 332min

So the route is (Depot, 14, 57, 51, 16, 48, 31, 50, 2, Depot).

4 Conclusions and Remarks

In this research work, we have tackled the challenging problem of optimizing the visit routes for a company's sales staff to efficiently serve its customers. We addressed the problem through the lens of the Vehicle Routing Problem with Time Windows (VRPTW) and considered various constraints such as working hours, customer frequency, distance matrix, and time windows.

To improve the efficiency of the sales staff, we divided the total customers' area into territories, which allowed us to solve smaller subproblems and optimize resource reuse by serving territories on different days. This strategy not only simplified the management of the system but also led to shorter service and travel times for salespeople, resulting in improved customer service.

To find optimal solutions, we employed linear programming with constraints and formulated the problem to include the working hours per day for each salesperson, customer frequencies, distances between customers, and appointment times. By leveraging heuristics and OR-TOOLS, we aimed to tackle the NP-hard nature of the problem and find near-optimal solutions.

It is essential to note that the quality of clustering had a significant impact on the routing solutions. Proper clustering ensured more efficient routes, avoiding longer travel times, higher operational costs, and potential customer dissatisfaction. Therefore, attention to proper clustering assignment is crucial for achieving desirable outcomes.

In conclusion, this research work contributes to the advancement of transport and logistics operations by providing efficient and cost-effective delivery systems. By addressing the challenges of the VRPTW and proposing innovative solutions, we aim to help the company optimize its sales staff routes, leading to improved customer service and overall efficiency.

The results achieved so far have been promising, and further refinement and exploration of optimization techniques may lead to even better outcomes. This research lays the groundwork for future improvements and opens the door for further studies in this domain, exploring more complex scenarios with multiple depots, heterogeneous vehicles, and additional constraints.

As the company continues to invest in industrial mathematics, we hope that the findings and proposed solutions from this research will positively impact their operations and contribute to their vision of becoming a world leader in transport solutions, construction equipment, and agricultural equipment.

Acknowledgement

We want to sincerely thank the Math-in organizations of Spain and Portugal for the Iberian Modeling Weeks in Mallorca. The events were enriching, fostering learning and knowledge exchange among experts and students in mathematical modeling. The collaborations in seeking innovative solutions to the transportation problem with time windows were incredibly valuable. Additionally, we thoroughly enjoyed the cultural exchanges with people from different countries. We express our gratitude to all the organizers and collaborators who made these unique experiences possible. We hope that these events continue to strengthen the scientific communities in Spain and Portugal. Thank you for these wonderful opportunities!

Overall, we are enthusiastic about the potential of this research to drive significant improvements in the field of transport and logistics, ultimately benefiting the company's goals and customer satisfaction. We extend our sincere gratitude to Professor Manuel Cruz for proposing this challenge and fostering research and innovation in the field of mathematical modeling.

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